

Technical Assessment: AI/ML Engineer

Company: Techsara Consulting Pvt. Ltd.

Role: AI/ML Engineer

Timebox: ~ 2 hours

1. Overview

This exercise is designed to drive conversation during the hiring process. At Techsara Consultancy, we value the ability to communicate technical change effectively. This assessment focuses on your ability to:

- Work with ML lifecycle tooling.
- Build and reason about reproducible ML workflows.
- Apply MLOps best practices regarding pipeline design, deployment, and observability.

Tooling Policy

You are **encouraged** to use a coding assistant (e.g., GitHub Copilot, ChatGPT, Claude, etc.) during this exercise.

- **Note:** In the subsequent interview, we will ask you to share your experience using the assistant: what worked, what didn't, and where it helped or hindered your progress.

2. Instructions

Step 1: Choose an Open-Source Project

Select an open-source project that allows you to make meaningful changes.

- **Fork/Clone:** Fork the repository and clone it locally. Work within a contained directory or branch for this task.
- **Selection Criteria:** The project must solve a real-world problem. A well-selected project that demonstrates significant impact will positively influence the evaluation process.
- **Constraint:** The project you choose **must be a new project** that does not currently exist in your portfolio/profile.

Step 2: Task Execution

You are required to improve the existing project (e.g., increasing model accuracy, creating a more effective preprocessing step, or swapping in a superior model) AND extend the workflow.

Core Task: Extend or Automate MLOps Lifecycle

Implement a small extension or automation relevant to the MLOps lifecycle. You may choose a backend framework if desired. Examples include:

- **Pipeline Steps:** Add a preprocessing, evaluation, or monitoring hook.
- **Versioning:** Automate versioning or artifact logging for an existing training script.
- **Deployment:** Wrap an existing model with a deployment-ready API (e.g., FastAPI + Docker).
- **Observability:** Add metrics tracking to a training or inference workflow.

Step 3: Mandatory Logic (The Conditional Step)

You must implement a logic gate in your pipeline that compares the new model against a production baseline (Model Registry).

- **The Logic:** The pipeline should only proceed to the deployment/packaging stage if the new model's metric (e.g., F1-score) is $\geq$ **the threshold** of the currently productionized baseline model.

3. Deliverables

Please package your submission to include the following:

A. Code

Your implementation (script, pipeline file, or notebook).

B. README.md

A short explanation (1–2 paragraphs) describing:

1. What you chose to implement and why.
2. How to run your code.
3. Any assumptions or limitations.

C. Reflection

A brief section (5–10 sentences) sharing your experience using the coding assistant, specifically addressing:

- Did it help you move faster?
- Did it generate incorrect or surprising suggestions?
- Where was it most or least useful?

4. Submission

1. Push your code to your forked repository.
2. Share the link to your repo/branch prior to the next interview session.

